

Macroeconomic Embedding Framework

Architecture, Data Sources, and Functional Synergy for E_macro

1. Core Embedding Definitions

E_local	Captures the physical geography, buildings, and infrastructure layout.
E_census	Captures the demographic profile, traits, and static wealth of the people who live there.
E_macro	Captures the movement of money, trade lines, and the overall momentum of the economy.

We can think of each embedding in this way: **E_local** is the stage, **E_census** is the actors, and **E_macro** is the actions.

2. Data Sources for E_macro (Comprised of A-F)

Source A: Wikipedia Introductory Corpora (Narrative Identity)

Programmatically query the English Wikimedia Enterprise API for the text introductions of all counties.

WHY IT IS A GOOD IDEA:

Text encoders possess pre-trained semantic associations that raw numbers obscure. Wikipedia captures the qualitative, historical transitions of an area. Phrases like "former rust-belt manufacturing center currently transitioning to a health-tech hub" or "exurban master-planned commuter community" provide immediate semantic distance within a sentence transformer (bge-m3). This breaks visual symmetries for locations that look identical from orbit or share identical population shapes.

Source B: BLS Quarterly Census of Employment and Wages (Industrial Core)

Hit the pre-sliced flat files via the BLS Open Data Access portal.

WHY IT IS A GOOD IDEA:

The QCEW records business employment and payroll at the workplace establishment level. It provides pre-calculated Location Quotients for every level of the North American Industry Classification System.

Source C: FRED Time-Series Slope Derivatives (Economic Velocity)

Query the Federal Reserve Economic Data API for county-level monthly and quarterly tracks: Local Area Unemployment Statistics and Gross Domestic Product by County. Rather than keeping the static values, calculate the rolling 3-year first derivative.

WHY IT IS A GOOD IDEA:

Census data is a trailing indicator that captures a static snapshot of population traits. FRED time-series slopes inject momentum into the data vector. A market with a high median household income whose job growth is slowing down will map to a completely different quadrant than an identical mid-income market experiencing sharp macroeconomic acceleration.

Source D: BTS FAF5 Experimental County Commodity Flows (Trade Gravity)

Ingest data from the Bureau of Transportation Statistics Freight Analysis Framework Experimental County-to-County Commodity Flow matrix. This tracks outbound and inbound freight tonnage and dollar value across Standard Classification of Transported Goods groupings.

WHY IT IS A GOOD IDEA:

This dataset maps the movement of physical goods across space. It cleanly distinguishes a logistical pass-through corridor or an industrial exporter from a pure consumer sink, a distinction that is completely invisible in demographic or visual map files.

Source E: IRS Statistics of Income Panel Data

Download the annual county-level data tables published by the IRS Statistics of Income division.

WHY IT IS A GOOD IDEA:

The Census records the volume of household income, but it obscures its nature. The IRS data allows you to compute the exact ratio of Form 1040 Schedule D (Capital Gains) + Qualified Dividends relative to Standard W-2 Wage Income. This factorized ratio isolates asset-rich, investment-driven markets (highly sensitive to monetary policy, equity performance, and interest rate adjustments) from pure labor-dependent markets (sensitive to corporate layoffs and manufacturing cuts).

Source F: USDA Economic Research Service (Structural Resilience Baseline)

Pull the deterministic, categorical County Typology Codes curated by the USDA Economic Research Service.

WHY IT IS A GOOD IDEA:

This dataset groups counties into mutually exclusive economic dependencies: Farming-dependent, Mining-dependent, Manufacturing-dependent, Government-dependent, or Recreation-dependent. For low-population or hyper-rural counties where online text data might be sparse or uninformative, appending these official structural labels directly to your input file provides a solid baseline. This anchors the text embedding model to a stable structural definition.

3. Functional Synergy: How These Fit Together

We can think of “the action” that can be captured by our embedding in three parts: Capital flow, economic velocity, and trade logistics.

1. Capital Flow: Tracking the "Fuel"

Capital flow tells the model what kind of money runs a local economy. If two neighborhoods look identical and have residents with the same average income, this pillar reveals if that income comes from stable corporate paychecks or volatile market investments.

Source E (IRS SOI Panel Data): This gives us the exact ratio of investment returns (capital gains/dividends) to standard W-2 wages. It tells the model whether a market's wealth is bound to local job survival or global Wall Street performance.

Source A (Wikipedia Introductory Corpora): This provides the qualitative context for that capital. It uses natural language to frame why capital is moving there—such as a shift from a historical manufacturing base toward a high-tech venture capital ecosystem.

2. Economic Velocity: Tracking the "Speed"

Economic velocity tells the model which direction a market is moving and how fast it is accelerating. It prevents downstream models from treating a booming economy and a collapsing economy identically just because their current static metrics happen to cross paths today.

Source C (FRED Slope Derivatives): By calculating the clean, deseasonalized trend-line slopes for local GDP and unemployment over a 3-year window, this source injects true momentum into the feature space.

Source B (BLS QCEW Location Quotients): This identifies the exact industrial engines driving that speed. It uses scale-invariant metrics to pinpoint whether a city's acceleration is powered by specialized white-collar tech hubs or blue-collar manufacturing centers.

3. Trade Logistics: Tracking the "Movement"

Trade logistics connects individual regions into a unified national network tensor. It tracks how physical commodities move across borders, revealing a market's structural dependencies and its vulnerability to supply chain shocks.

Source D (BTS FAF5 Commodity Flows): This tracks the physical mass—the inbound and outbound freight tonnage and dollar value—crossing regional lines. It isolates critical transport hubs and industrial exporters from simple consumer endpoints.

Source F (USDA Economic Research Service Typologies): This anchors the long-tail and rural markets to their fundamental economic identities (e.g., farming-dependent vs. mining-dependent). It establishes a baseline for how these regions absorb broader supply chain pressures.